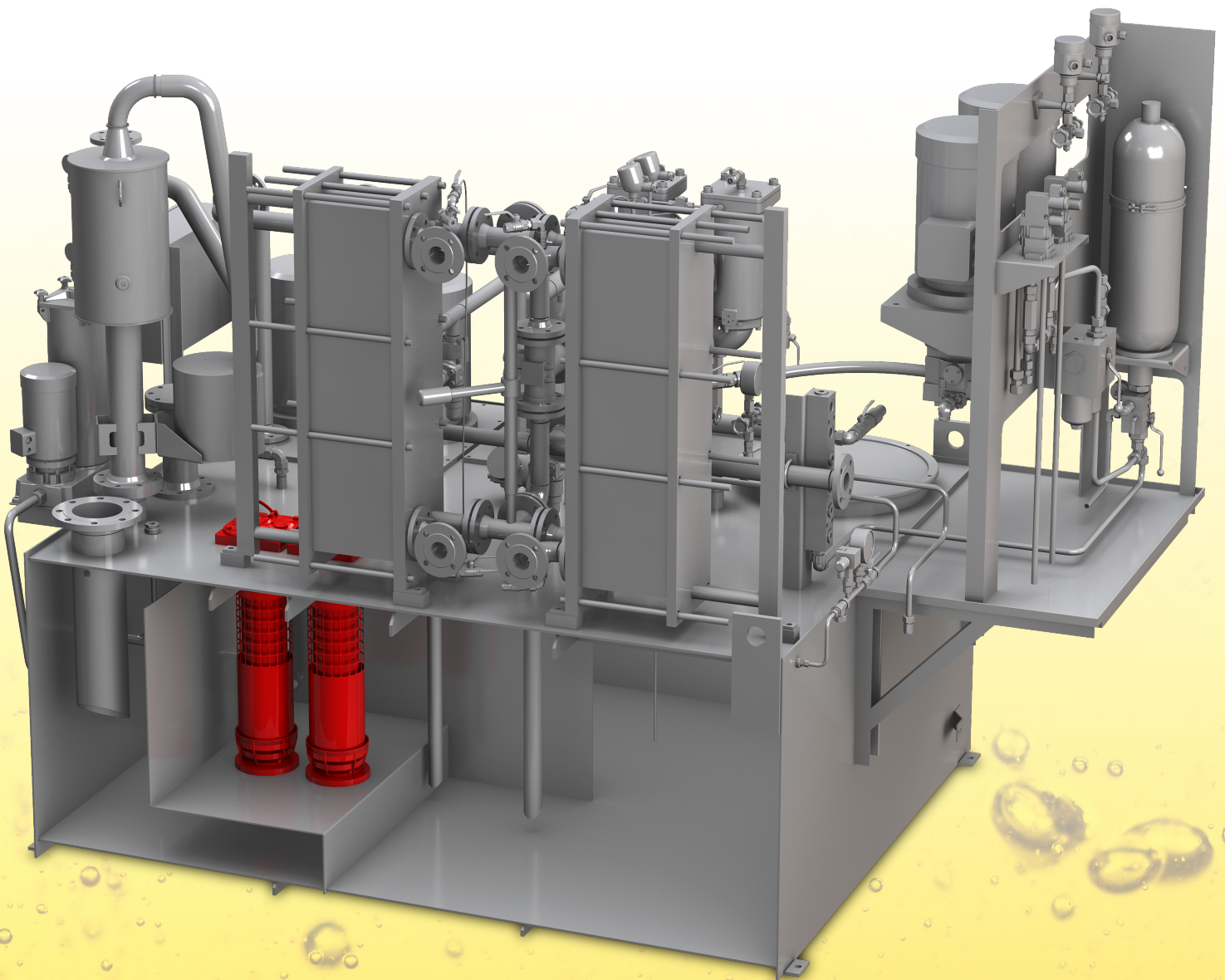


Tank Concept with Air-X Technology

for low-pressure systems with oils < 50 cSt



Tank Concept with Air-X Technology

Challenge

A breakdown of the lubrication film in machine elements may result in major damage to the machine itself and furthermore to production downtimes. Lubrication systems are therefore used to ensure that bearings, gearboxes, mechanical seals etc. such as those in turbines, process compressors and process pumps are well lubricated.

Lubricating oil systems also channel away abrasion debris and friction heat. Even air that may have entered the system via the mechanical seals, for example, needs to be removed with almost zero residue. The entry of air into the system is also deliberately increased by oil mist separators. The negative pressure generated in the tank continues up to the mechanical seals and ensures that air is taken in there rather than oil being released to the outside.

Current state of technology

Air bubbles, which may be finely distributed in the oil, rise to the surface in the flowing oil, where they are then "separated". This process takes time, so conventional lubricating oil tanks are designed for the journey from return line → oil inlet and oil outlet → suction line to take six to eight minutes. Special tank installations channel the oil through the tank.

Optimum air separation through innovation

Experiments have shown that using the new HYDAC degassing components, which are presented below, can reduce the dwell time of 6 – 8 min, which is typical for the application, to 2 min, depending on how much air enters via the return lines.

For original tank sizes from 100 to 20,000 litres, this means enormous financial savings, a greatly reduced environmental impact, a drastic reduction in the amount of floor space needed and major weight reductions in the steel construction.

Customer Advantages

- **Large cost reductions thanks to reduced material use**
(steel, oil (both at initial start-up and during every oil change), smaller fluid conditioning units)
- **Enormous reduction in weight**
meaning that costs can also be saved in the design of the building
- **Reduced logistics expenditure**
thanks to lighter and smaller tank
- **Increased sustainability**
thanks to lower oil quantity

Note

The information in this brochure relates to the operating conditions and applications described. For applications and / or operating conditions not described, please contact the relevant technical department. Subject to technical modifications.

HYDAC Product Series Degassing Unit RKB

Function in detail

The special degassing concept of the RKB product series enables the quantity of air in the oil to be reduced quickly.

The degassing components are installed close to the return flow. To prevent an undesired backlog from occurring there, the components are designed to be operated with very low pressure losses of roughly 5 mbar at the operating point.

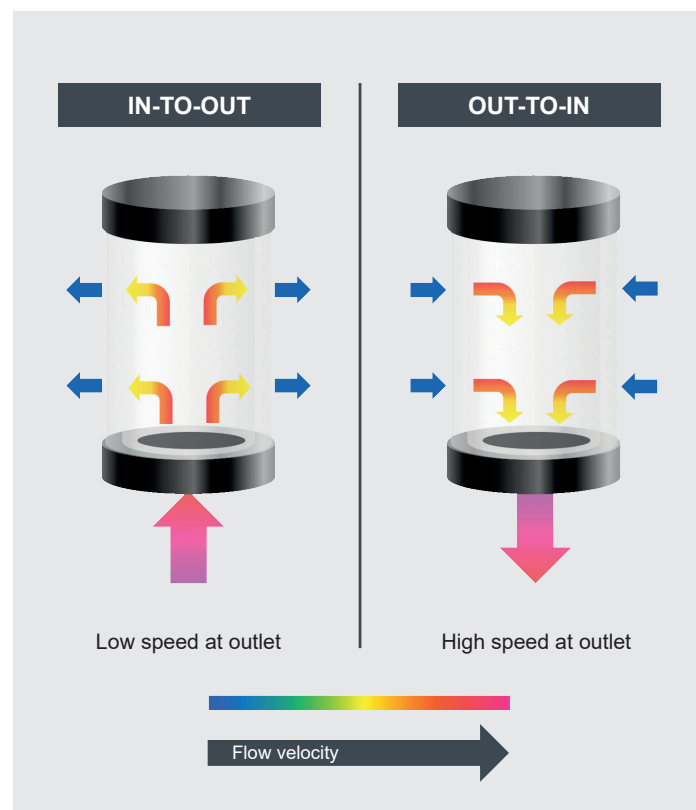
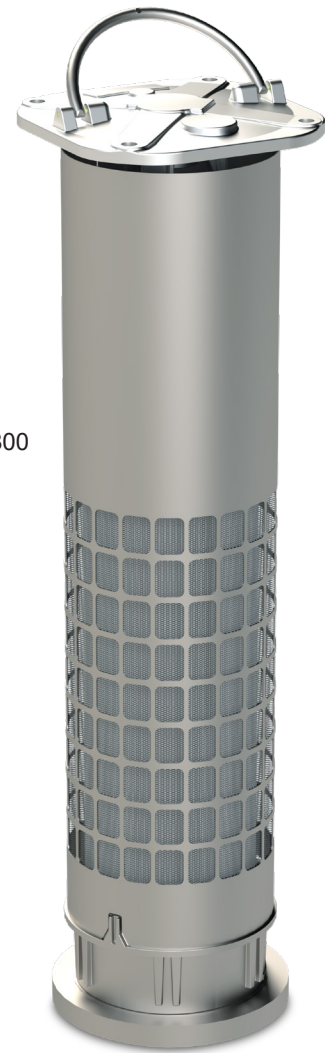
Example: RKB 800

Flow direction

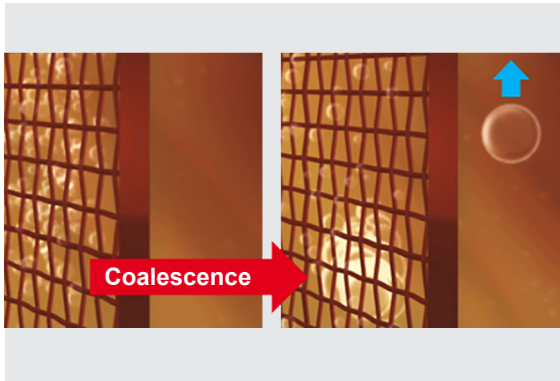
In all the degassing components in the series, flow passes from the inside to the outside.

One of the important properties of this flow direction is that it calms the flow towards the tank, as well as entrapping coarse dirt on the flow side when the inner Air-X unit is changed. Because of the larger outlet flow cross section, the oil flows out of the degassing unit much slower. The air bubbles in the oil therefore have more time to rise.

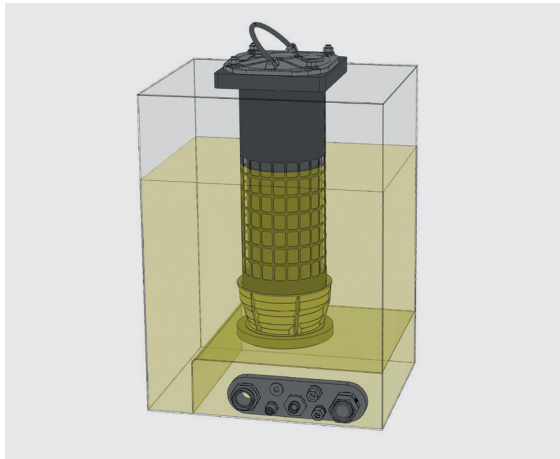
The even and slow exit flow of the oil also results in a calm oil level in the tank. There is no "splashing" in the tank, which could cause air to enter back into the fluid.



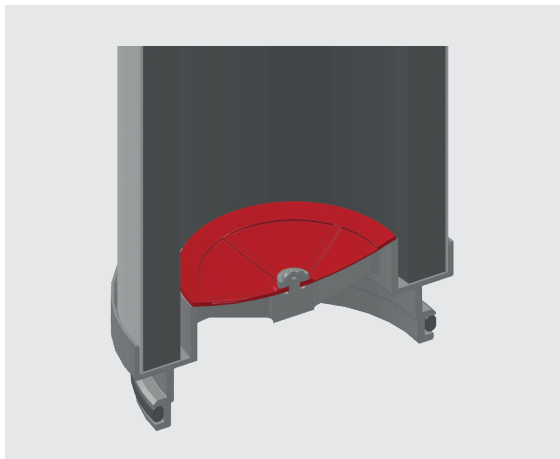
Tank Concept with Air-X Technology



Coalescence effect at the metal grate



RKB degassing unit as tank installation



Anti-drain valve

Coalescence

From a physics perspective, the time an air bubble takes to rise in the oil is partly dependent on the bubble's diameter. The relationship is quadratic. A bubble with a diameter that is smaller by a factor of 10 will have a rising time that is longer by a factor of 100.

Smaller bubbles therefore have a higher risk of being carried away by the flow. If the oil flows through the filter medium, the fine filter material makes the air bubbles smaller.

The innovative products with the patented "window outflow" with metal mesh cause smaller bubbles to merge to form larger ones. This merging is referred to as coalescence. The resulting large bubbles rise much faster and the separation of air is significantly improved.

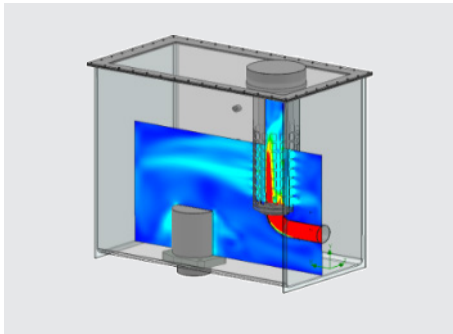
Incident flow

In the tank systems under discussion here, fluid flows to the degassing components from below, so part of the oil flow rate can flow out near the surface, through the windows in the housing. This reduces the distance that air bubbles have to cover to rise to the surface to a minimum and further increases the system's air separation.

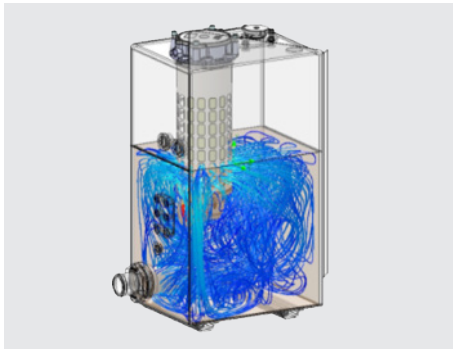
The RKB degassing components are perfect for large flow rates. They are mounted in the intermediate chamber that is formed in the bottom part of the tank. The oil therefore flows into the degassing unit from below and then into the tank via the riser tube. Because of the intermediate chamber, any design is possible for the return port connections and several components can be used without splitting up the return lines. The optimal flow conditions in the chamber guarantee optimum air separation.

If the assembly is to be used as a degassing unit to retain coarse dirt, the patented anti-drain valve can be used optionally. This prevents oil from returning from the contaminated side to the clean side during element change and the dirt remains trapped in the element.

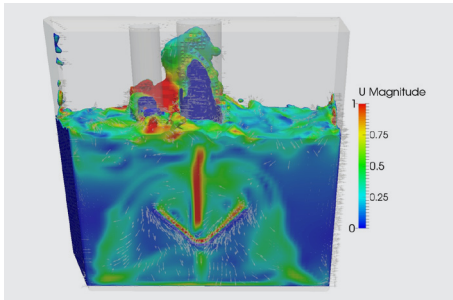
The Development of the Tank Concept



Flow



Degassing

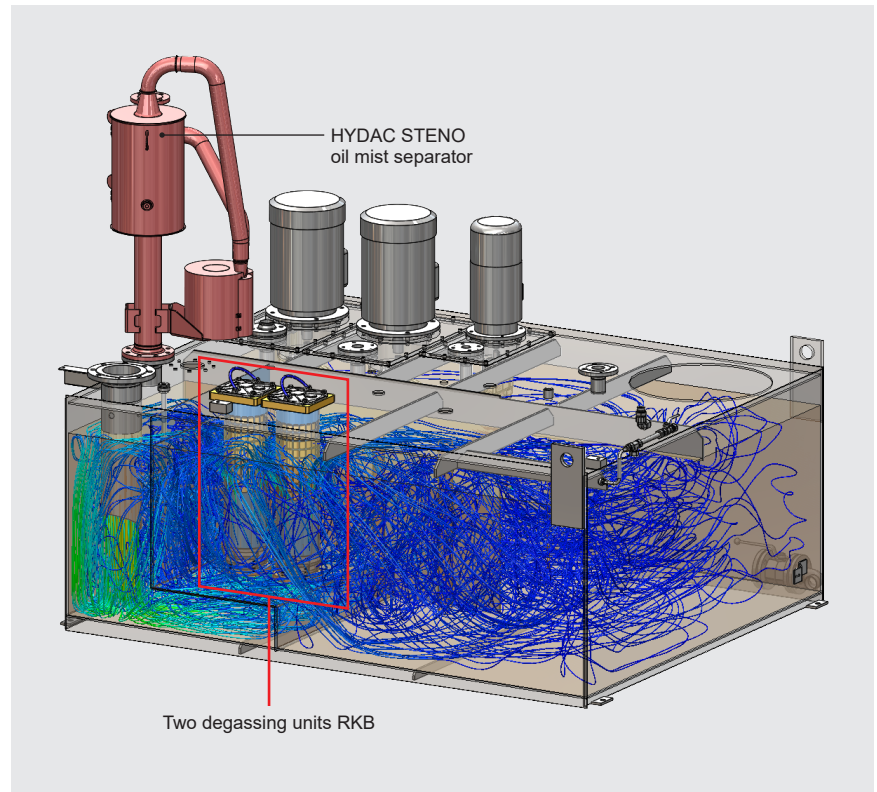


Multi-phase flow

Simulation

HYDAC has the capacity and the necessary experience to describe hydraulic and lubricating oil tanks in detail. In addition to calculating flows, two-phase flows and complex multi-phase flows can be implemented to observe the surface effects of fluids (surge movements).

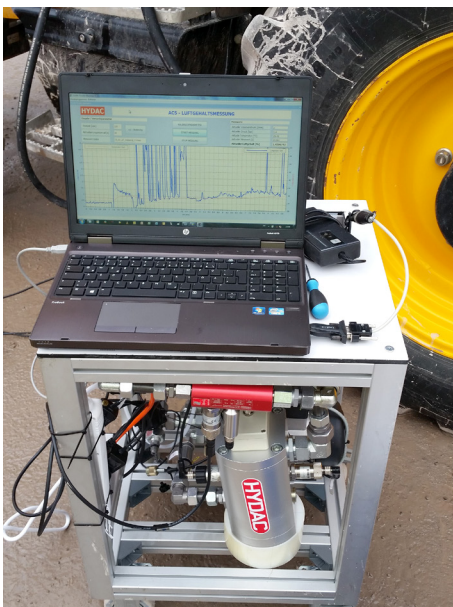
Here is an example of two-phase simulation of the tank concept presented below:



Measurement

To perform robust measurements, HYDAC has developed a sensor for measuring the air content. This sensor makes it possible to measure vertical and longitudinal profiles in a tank and to determine the air distribution with great accuracy.

There are a series of test facilities at several HYDAC sites where the findings of the simulation can be verified. The measurement sensor is mobile and can also be used in field tests. This can be useful first to determine the initial situation before the degassing units are used and second to test the system after successful modification with the HYDAC products.

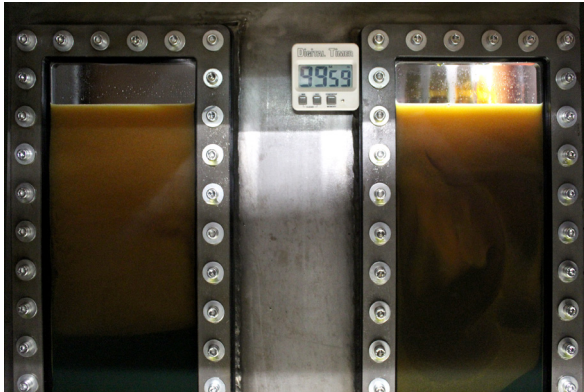


ACS air content sensor

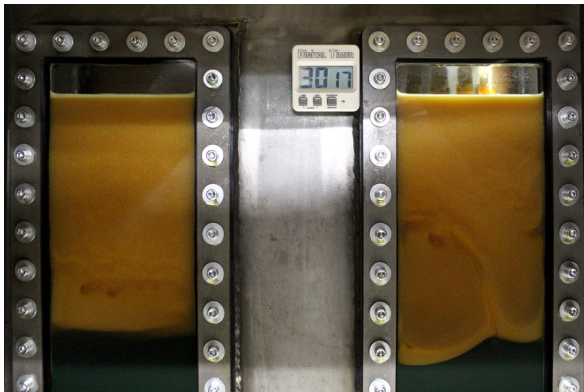
Tank Concept with Degassing Unit

Different air contents in the tank

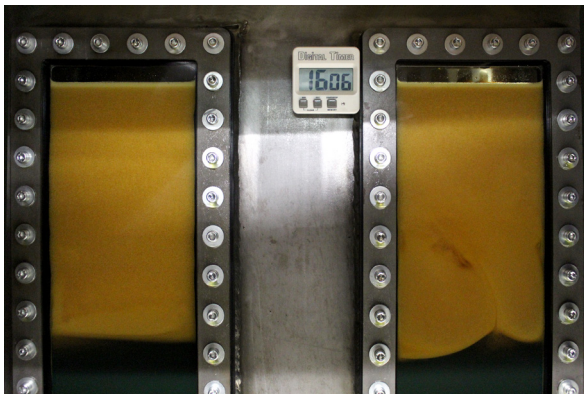
Air content = 2 vol. %



Air content = 5 vol. %



Air content = 10 vol. %

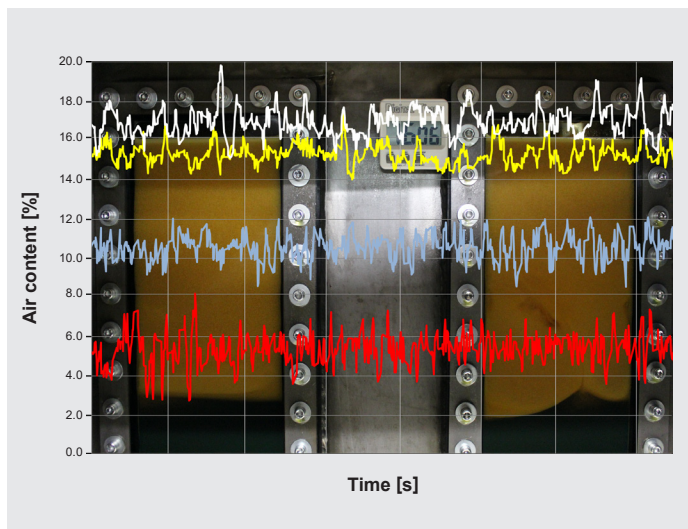


Measurement results, test setup

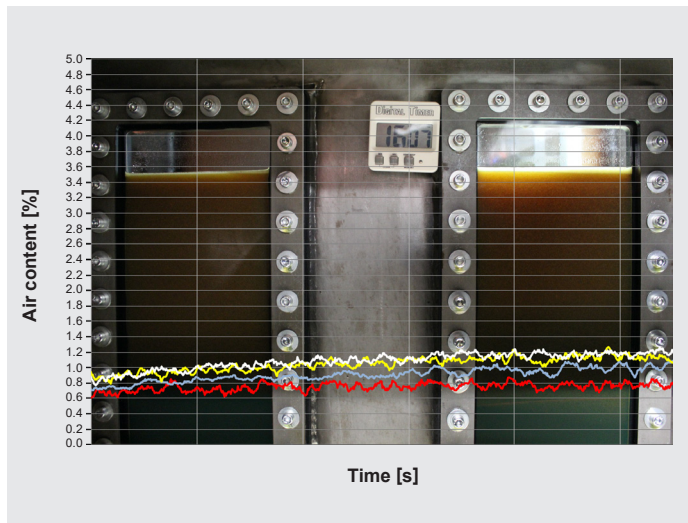
Tests at the HYDAC FluidCareCenter have shown that even high air content in the oil, distributed in fine droplets, can be sufficiently removed in the degassing stages. The following figures show examples of tests with various air concentrations in the oil at 200 l/min and a tank dwell time of two minutes in the return flow and, for comparison, the residual air content after the degassing stages.

The colour-coding for before and after is the same in all cases:

Before: different air concentrations in the oil

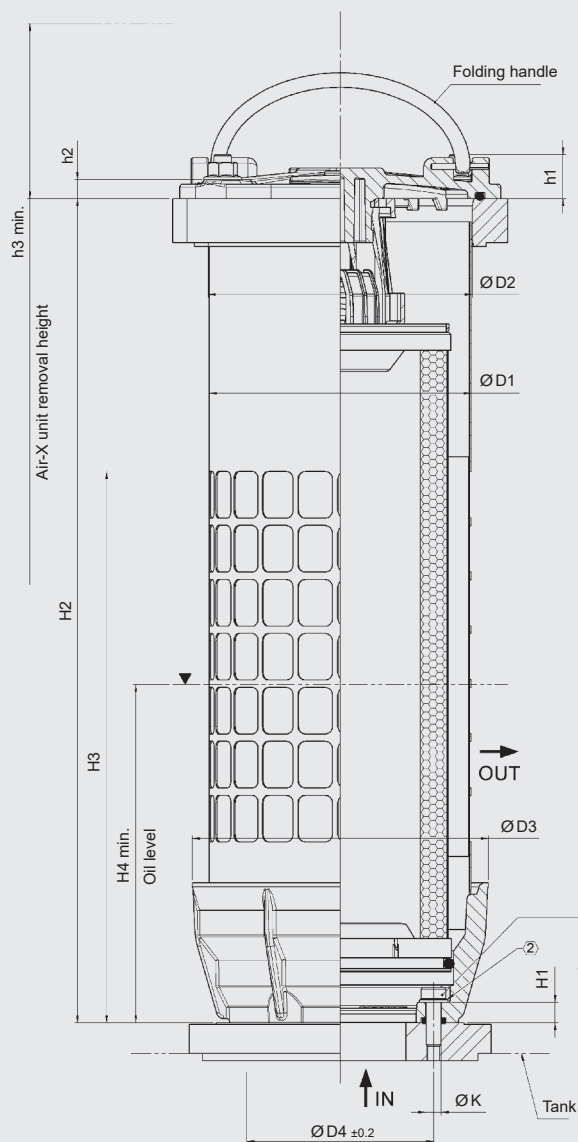


After: air concentrations after the degassing stages

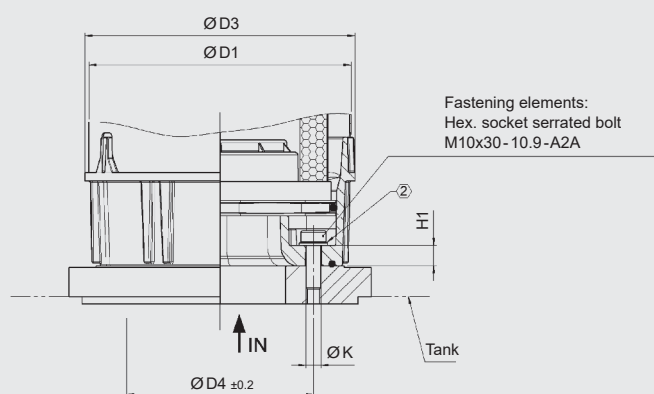


HYDAC Product Series

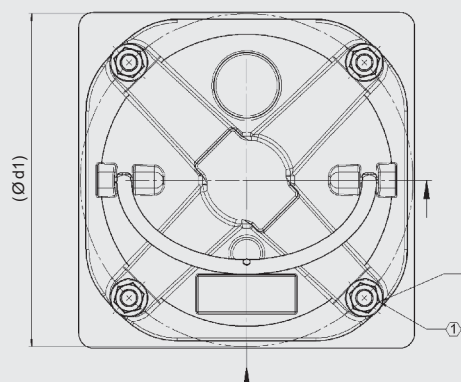
Degassing Unit RKB



Degassing unit with poka yoke
(tank flange version C; I)



Fastening elements:
Hex. socket serrated bolt
M10x30 - 10.9 - A2A
Recommended torque 40 +5 Nm
(when specified standard part is used
with steel as material for the flange)

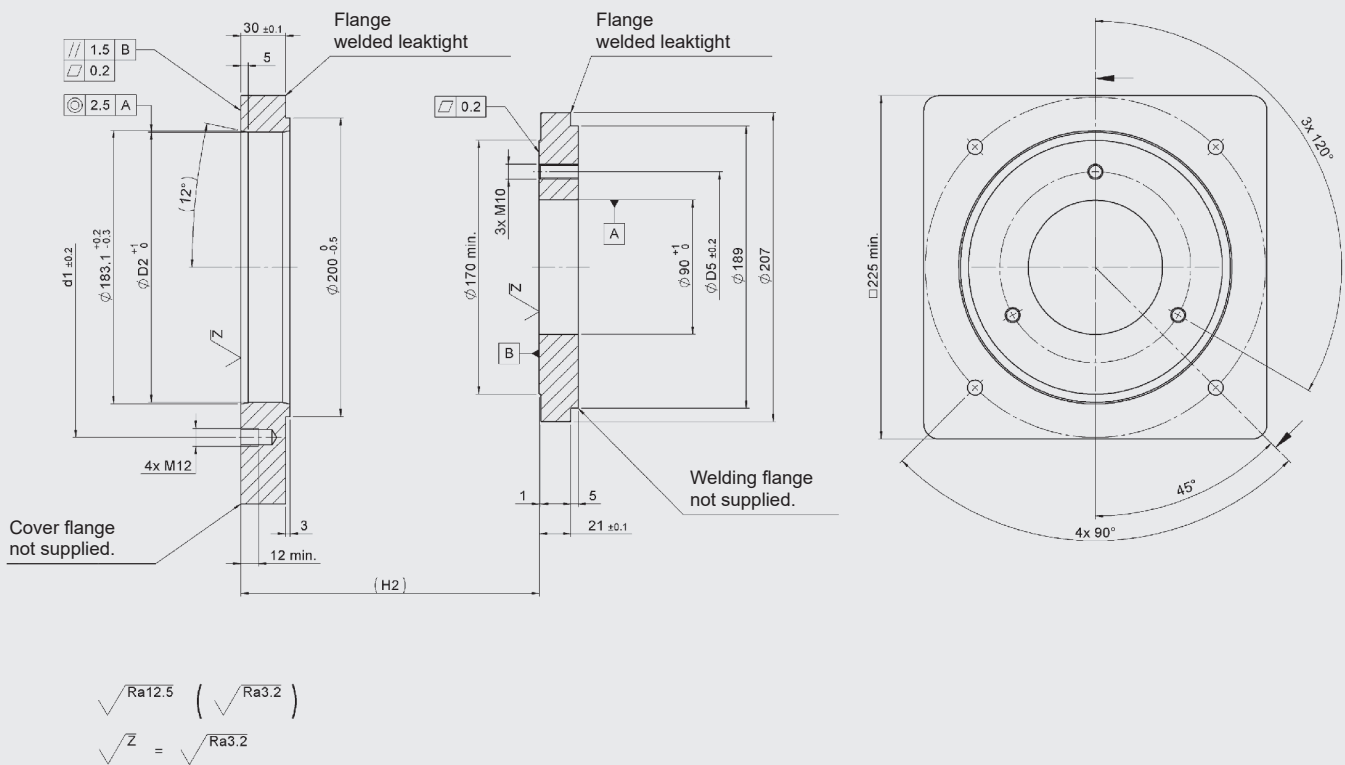


Fastening elements, e.g.:
Hexagon nut with collar DIN EN 1661-M12
Stud DIN938 - M12x30 - 8.8

Type	Tube version	Tank flange	H1	H2	H3	H4	h1	h2	h3	ØD1	ØD2	ØD3	ØD4	ØK	Ød1	Weight [kg]
RKB 0600	R	B; E	14	565	378	232	30	13	485	179.5	181	203	128	10.5	228	8.8
		C; I										185				8.7
RKB 0800	R	B; E		758	452	269			678			203				11.6
		C; I														185
RKB 1200	R	B; E		912	600	343			832			203				14.0
		C; I														185

Specifications for the tank flange

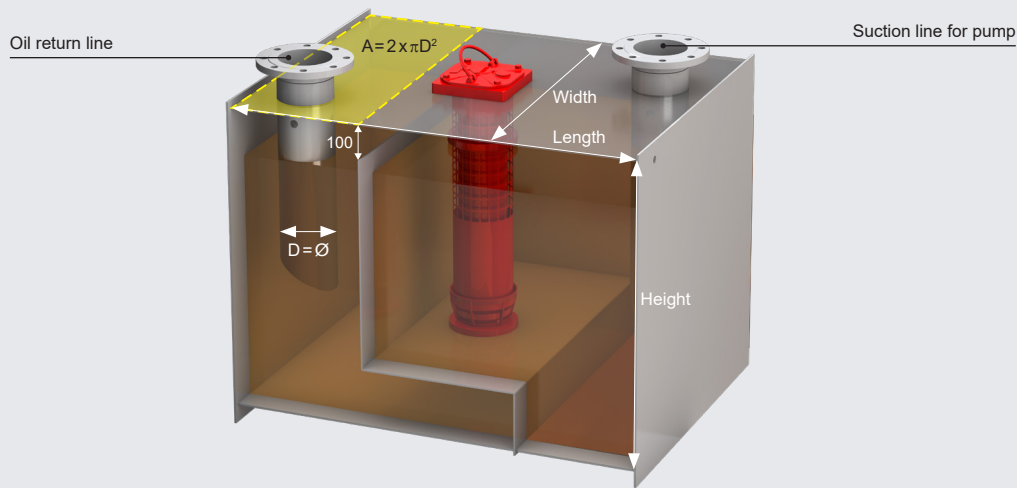
- At the degassing unit mounting interface, the tank flange should have a maximum flatness of 0.2 mm and maximum roughness of Ra 3.2 µm.
- In addition, the mounting interface should be free from damage and scratches.
- The fixing holes of the cover flange must be blind, or stud bolts with threadlocker must be used to fix the degassing unit.
As an alternative, the cover flange can be continuously welded from the inside.
- Both the tank sheet metal and the degassing unit mounting flange must be sufficiently robust so that neither deform when the seal is compressed during tightening.



The cover flange must be aligned to the welding flange with a welding gauge.

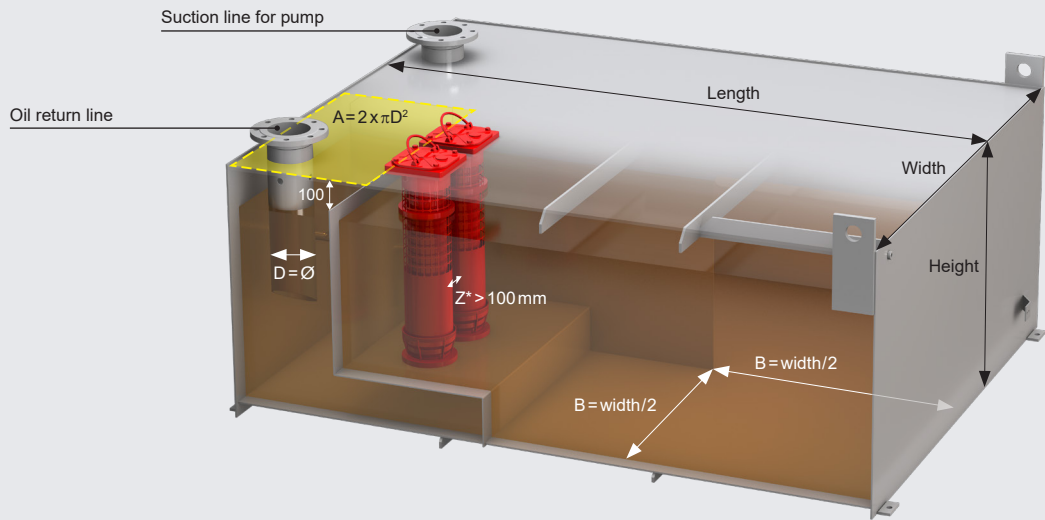
Degassing unit assembly		Anti-drain valve	Q _{max} [l/min]	Spare part AIR-X degassing unit	
Part no.	Model code			Part no.	Model code
1901305	RKB 1200 WPI 200 V X B R X C N VXX 1/SAL	–	300	1901338	WPI-0200-152-XD31-Q-N-K/SAL
1901327	RKB 1200 WPI 200 V X B R V C N VXX 1/SAL	x	300	1901339	WPI-0200-152-XF31-Q-N-K/SAL
1901306	RKB 0800 WPI 200 V X B R X C N VXX 1/SAL	–	200	1901336	WPI-0200-152-XD25-Q-N-K/SAL
1901328	RKB 0800 WPI 200 V X B R V C N VXX 1/SAL	x	200	1901337	WPI-0200-152-XF25-Q-N-K/SAL
1901307	RKB 0600 WPI 200 V X B R X C N VXX 1/SAL	–	150	1901334	WPI-0200-152-XD18-Q-N-K/SAL
1901329	RKB 0600 WPI 200 V X B R V C N VXX 1/SAL	x	150	1901335	WPI-0200-152-XF18-Q-N-K/SAL

Design recommendations for tank volumes up to 1,600 litres



Tank		Tank volume [l]		Tank dimensions [m]			Volume [l]		Degassing unit	Air intake 0–15 %			Air intake <5 %		
Size	from	to	Width	Length	Height	Oil level	Tank	Oil	Type	Flow rate [l/min]	Time [min]	Number of units	Flow rate [l/min]	Time [min]	Number of units
Small	100	1,600	0.5	1.0	0.8	0.6	400	300	RKB 600	150	2	1	150	2	1
			0.5	1.5	0.8	0.6	600	450		225	2	2	225	2	1
			0.5	2.0	0.8	0.6	800	600		300	2	2	300	2	1
			0.8	2.0	0.8	0.6	1,280	960		480	2	3	480	2	2
			0.8	2.5	0.8	0.6	1,600	1,200		600	2	4	600	2	2

Design recommendations for tank volumes up to 27,000 litres



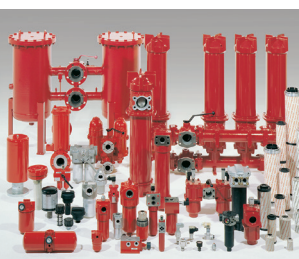
* Minimum distance Z between each housing of a degassing unit 100 mm

Tank		Tank volume [l]		Tank dimensions [m]			Volume [l]		Degassing unit	Air intake 0–15 %			Air intake <5 %		
Size	from	to	Width	Length	Height	Oil level	Tank	Oil	Type	Flow rate [l/min]	Time [min]	Number of units	Flow rate [l/min]	Time [min]	Number of units
Medium	1,500	9,000	1.0	1.5	1.0	0.7	1,500	1,050	RKB 800	525	2	3	525	2	0
			1.0	2.0	1.0	0.7	2,000	1,400		800	2	4	800	2	2
			1.0	2.5	1.0	0.7	2,500	1,750		875	2	4	875	2	2
			1.0	3.0	1.0	0.7	3,000	2,100		1,050	2	5	1,050	2	3
			2.0	2.0	1.0	0.7	4,000	2,800		1,400	2	7	1,400	2	4
			2.0	2.5	1.0	0.7	5,000	3,500		1,750	2	9	1,750	2	4
			2.0	3.0	1.0	0.7	6,000	4,200		2,100	2	11	2,100	2	5
			2.0	4.0	1.0	0.7	8,000	5,600		2,800	2	14	2,800	2	7
Medium to large	9,000	27,000	2.0	3.0	1.5	1.2	9,000	7,200	RKB 1200	3,600	2	12	3,600	2	6
			2.0	4.0	1.5	1.2	12,000	9,600		4,800	2	16	4,800	2	8

Medium to large tanks with more than 16 degassing units and/or a tank volume of more than 27,000 litres are possible with customised dimensioning on request.



Accumulator Technology 30,000



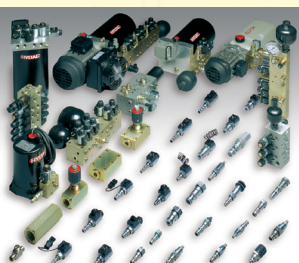
Filter Technology 70,000



Process Technology 77,000



Filter Systems 79,000



Compact Hydraulics 53,000



Accessories 61,000

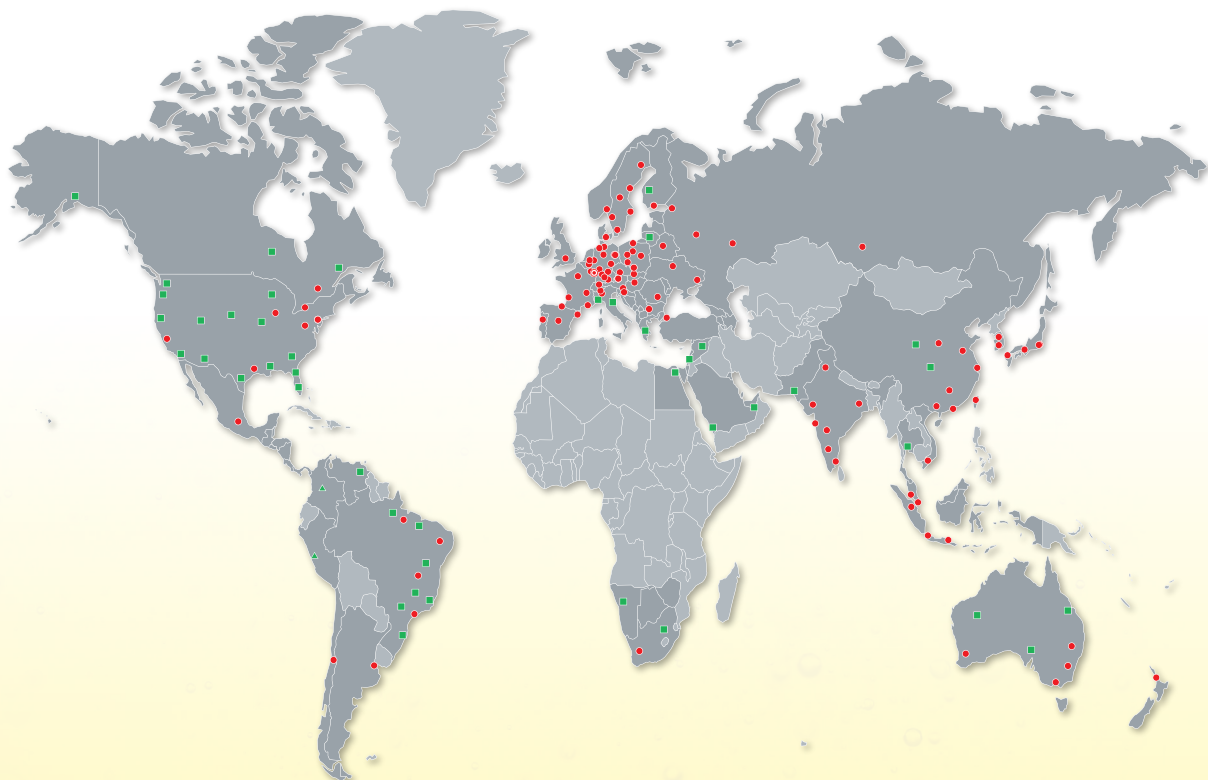


Electronics 180,000



Cooling Systems 57,000

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